

Brazilian E-Commerce Performance Analysis

Revenue, customer, delivery, seller and product insights from the Olist public dataset

Dataset: Brazilian E-Commerce Public Dataset by Olist (Kaggle) **Period:** Oct 2016 – Aug 2018 **Tools:** Python (pandas, NumPy), Google Colab

R\$15.2M Revenue (delivered orders)	95,118 Orders analysed	92,071 Unique customers	91.9% Delivered on time
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Executive summary

This project analyses about two years of marketplace activity from Olist, a Brazilian e-commerce platform that connects small sellers with customers nationwide. Nine raw tables were cleaned using a bronze / silver / gold layered approach, joined into consolidated datasets and used to answer business questions about revenue, customers, delivery, sellers and products. All figures cover delivered orders with a known product category.

Key findings

- **Fast growth.** Revenue reached R\$6.81M in 2017 and R\$8.37M in 2018 with data only up to August. Comparing January to August of both years, 2018 revenue is about 2.4x that of 2017. November 2017 was the strongest month at R\$1.14M, up 55% on October.
- **Growth has levelled off.** Since January 2018 monthly revenue has stayed in a R\$0.95M to R\$1.12M band instead of continuing to climb.
- **Retention is the biggest weakness.** 97% of customers placed exactly one order. Only 213 customers bought three to five times and just 8 bought six or more.
- **Sales are geographically concentrated.** São Paulo alone accounts for 37% of revenue, and São Paulo, Rio de Janeiro and Minas Gerais together for 63%.
- **Logistics is solid.** 91.9% of orders arrived before the estimated delivery date; 8.1% (7,709 orders) were late.
- **Quality varies widely.** Average review scores range from 4.67 (music CDs/DVDs) down to 2.50 (insurance and services), and seller ratings span the full 1 to 5 scale.

1. Business context and objectives

Management asked for a data-driven view of marketplace performance. The analysis was structured around these questions:

- **Revenue:** How does revenue evolve over time and how is it distributed across states, categories and customers?
- **Customers:** How can the customer base be segmented by loyalty?
- **Delivery:** How reliably are orders delivered by the estimated date?
- **Sellers:** Which sellers lead in revenue, order volume and customer satisfaction?
- **Products:** Which categories lead on volume, and which receive the worst reviews?

2. Data and methodology

2.1 Layered data pipeline

The notebook follows a medallion-style structure. The bronze layer holds the raw Kaggle CSVs (customers, geolocation, order items, payments, reviews, orders, products, sellers and a category translation table). The silver layer holds cleaned, de-duplicated tables. The gold layer holds the consolidated datasets and metrics used for analysis.

Table	Issue found	Treatment
Customers	Inconsistent city text; no duplicate customer IDs	Standardised city names to title case.
Items	9,803 orders appear on more than one row (up to 21 rows)	Investigated sample orders, then kept one row per order ID.
Payments	Several payment rows per order (up to 29)	Grouped by order and payment type, then pivoted into one column per payment method.
Reviews	87,656 missing titles and 58,247 missing comments; 789 review IDs repeated	Dropped the sparse title column; scores are complete. Repeats resolved after joining, keeping the latest review per order.
Orders	160 missing approval, 1,783 missing carrier and 2,965 missing delivery dates	Dropped approval and carrier columns; kept delivery date for delivery analysis; date fields converted to datetime.
Products	610 products with no category; text underscores in names	Excluded products without a category; standardised names.
Sellers	Inconsistent city text	Standardised city names.

2.2 Consolidated model

Orders were joined with items, products, customers and payments, then filtered to delivered orders. Payment types were unpivoted to calculate revenue as the amount paid by customers (which includes freight). A separate seller-level dataset joined items, sellers, orders, reviews and products.

3. Analysis and findings

3.1 Revenue over time

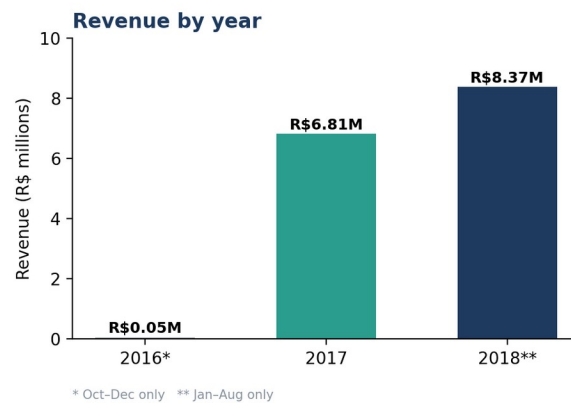


Figure 1. Annual revenue. 2016 and 2018 cover partial years.

Total revenue reached R\$15.2M. 2017 delivered R\$6.81M, and 2018 had already reached R\$8.37M by August, 23% above the full previous year. 2016 (R\$46K, October to December only) is not comparable.

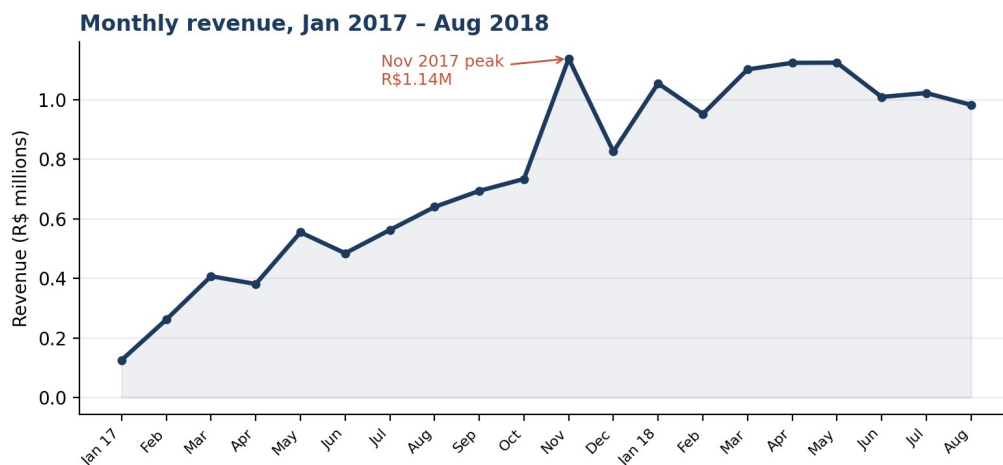


Figure 2. Monthly revenue, January 2017 to August 2018.

- **2017 ramp-up:** revenue climbed from R\$126K in January to R\$734K in October.
- **November 2017 spike:** R\$1.14M, up 55% on October. The timing is consistent with Black Friday promotions, although the dataset does not confirm the cause.
- **2018 plateau:** monthly revenue moved between R\$0.95M and R\$1.12M from January to August.

Note: October and December 2016 hold only a handful of orders (December: R\$19.62) and reflect the start of the dataset, not real trading levels.

3.2 Revenue by state and category

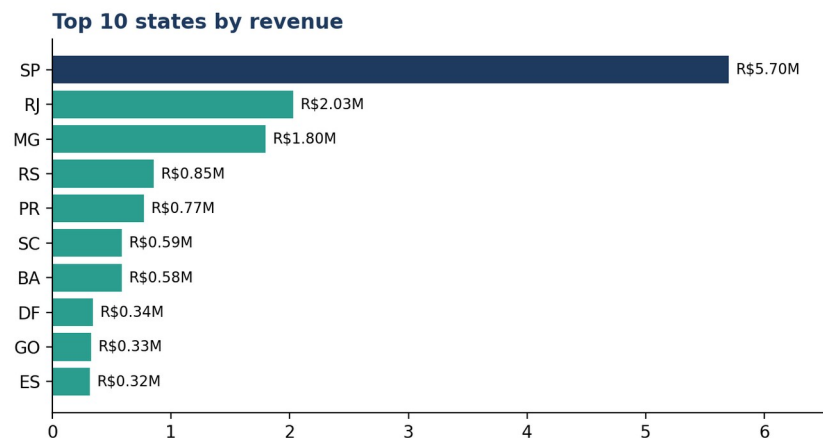


Figure 3. Top 10 states by revenue.

São Paulo generated R\$5.70M (37.4% of revenue), followed by Rio de Janeiro (R\$2.03M, 13.3%) and Minas Gerais (R\$1.80M, 11.8%). The top three states account for 63% of revenue and the top five for 73%. Northern states contribute very little: Roraima (R\$8.7K), Amapá (R\$16.1K) and Acre (R\$19.5K).

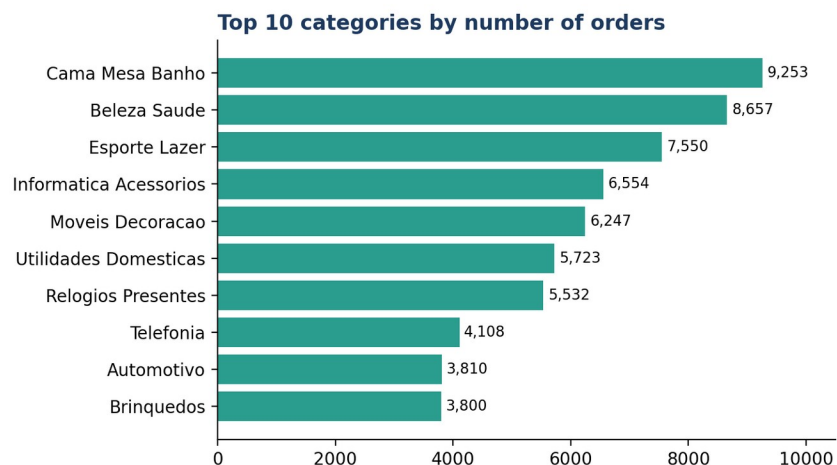


Figure 4. Top 10 product categories by number of orders.

Bed, bath and table leads with 9,253 orders, followed by health and beauty (8,657) and sports and leisure (7,550). By revenue, home utilities (R\$758K) and telephony (R\$381K) are among the biggest contributors.

3.3 Customer segmentation

Customers were segmented by purchase frequency: Occasional (1-2 orders), Loyal (3-5 orders) and VIP (6 or more).

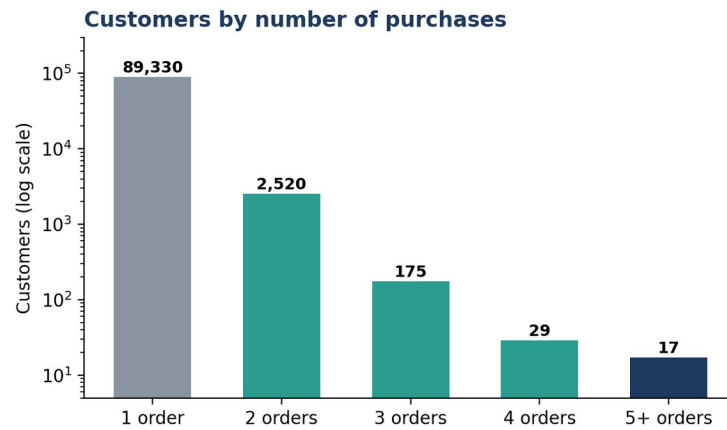


Figure 5. Customers by number of purchases (log scale).

Segment	Definition	Customers	Share
Occasional	1-2 orders	91,850	99.8%
Loyal	3-5 orders	213	0.2%
VIP	6+ orders	8	<0.01%

Of 92,071 customers, 89,330 (97.0%) bought only once, and the most loyal customer placed 14 orders. Acquisition is working, but retention is not.

3.4 Delivery performance

Delivery was measured against the estimated delivery date. Of 95,118 delivered orders, 87,409 (91.9%) arrived on time and 7,709 (8.1%) arrived late. Reliable logistics is a strength worth protecting, and late deliveries are a natural next area to link to review scores.

3.5 Seller performance

Sellers were ranked on revenue, order count and average review score (3,020 sellers with reviews).

- **Revenue is concentrated at the top.** The five highest earners generated between R\$173K and R\$239K each (about R\$1.03M combined), while the smallest sellers earned only a few reais.
- **Volume leaders.** The busiest seller handled 1,819 orders, followed by 1,736 and 1,696. One seller (ID ending 493884) ranks third in revenue and second in orders.
- **Quality is uneven.** Average scores span 1.0 to 5.0. Many perfect scores probably come from sellers with very few orders, so score should be read together with volume.

3.6 Product analysis

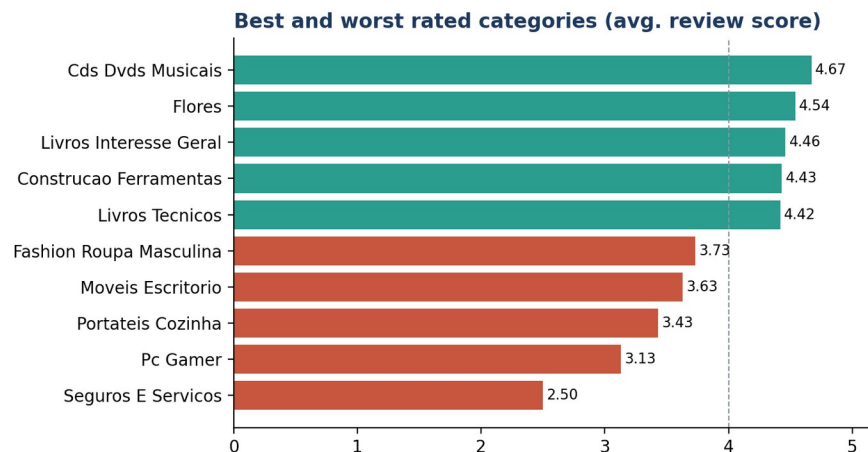


Figure 6. Highest and lowest rated product categories.

Simple, low-risk categories such as music CDs/DVDs (4.67), flowers (4.54) and books (4.46) earn the best reviews. The weakest are insurance and services (2.50), gaming PCs (3.13) and kitchen appliances (3.43). Technically complex products tend to disappoint more, and small categories can swing sharply on a few reviews. Month-over-month category growth was also computed, but percentage changes on small categories are too volatile to rank reliably without smoothing.

4. Recommendations

- **Invest in retention.** With 97% of customers buying once, even a small lift in repeat purchases adds revenue without extra acquisition cost. Options include post-purchase emails, loyalty rewards and cross-sell in high-volume categories.
- **Plan for seasonality.** Prepare stock, logistics and marketing for the November peak, and test campaigns to lift flatter months.
- **Address the plateau and concentration.** Flat 2018 revenue and a 63% reliance on three states suggest saturation in core markets; explore under-served regions.
- **Protect delivery reliability.** Keep the 92% on-time rate as volume grows and investigate the late 8%.
- **Manage seller and category quality.** Set service standards and monitor categories below 3.5 average review, especially technical products such as gaming PCs and appliances.

5. Limitations and next steps

- **De-duplication method.** Items were reduced to one row per order ID. Because each row in Olist is one unit purchased, multi-item orders lose their extra items. Revenue is taken from payments so it is largely unaffected, but item-level metrics and seller or category attribution may be understated.
- **Average order value.** Dividing total payment revenue by orders gives about R\$160. The notebook's R\$125 uses a single item price per order and excludes freight, so it understates the true figure.

- **Freight in customer spend.** Payment values already include freight, so adding freight again in the customer spend table double counts it.
- **Recency reference date.** Recency was measured against today's date, which makes every customer look years inactive. Using the last date in the dataset as the reference would enable a proper RFM model and the At Risk segment.
- **Partial periods and scope.** 2016 and 2018 are incomplete years, and figures cover delivered orders only.
- **Causes are not proven.** Seasonality, geography and quality patterns are observed associations. Suggested next steps: delivery time versus review score, seller concentration (Pareto), smoothed category growth and an interactive Power BI or Tableau dashboard.